QR Code for report verification

Case ID	:	2600135672	Sample Type	:	BLOOD IN CCF DNA TUBE
Name	:	MR. M KUMAR (RPR2604290001)	Date & Time Collected	:	29-Apr-2026 02:32 AM
Sex/Age	:	Male/47 Years	Date & Time Received	:	01-May-2026 05:12 PM
Bill. Loc.	:	OncQuest Laboratories Ltd., New Delhi	Date & Time Reported	:	14-May-2026 03:26 PM
Ref. By	:		Report Version	:	1
Indication	:				

LiquidSeq- Colorectal Cancer Panel-Liquid Biopsy Assay

Clinical Indication:

Moderately Differentiated Adenocarcinoma; Colonic Biopsy

Negative

(No Clinically significant variants detected)

Key Findings:

Gene & Transcript	Variant	Exon	Coverage / VAF	Classification	Relevant Biomarker
NA					

Assay Biomarker	Result	Interpretation
Microsatellite Instability (MSI)	Stable	Stable Microsatellite detected

**Genetic test results are reported based on the recommendations of AMP-ASCO-CAP guidelines.*

Test Description:

LiquidSeq- Colorectal Cancer panel is an NGS panel based on molecular tagging technical method that tested to detect low MAFs mutation from plasma samples.

Variant Classification as per AMP-ASCO-CAP Guidelines:

Variant	A change in a gene. This could be disease causing (pathogenic) or not disease causing (benign).
Tier I: Variants of Strong Clinical Significance	**Level A**, biomarkers that predict response or resistance to US FDA-approved therapies for a specific type of tumor or have been included in professional guidelines as therapeutic, diagnostic, and/or prognostic biomarkers for specific types of tumors; **Level B**, biomarkers that predict response or resistance to a therapy based on well-powered studies with consensus from experts in the field, or have diagnostic and/or prognostic significance of certain diseases based on well-powered studies with expert consensus.
Tier II: Variants of Potential Clinical Significance	**Level C**, biomarkers that predict response or resistance to therapies approved by FDA or professional societies for a different tumor type (ie, off-label use of a drug), serve as inclusion criteria for clinical trials, or have diagnostic and/or prognostic significance based on the results of multiple small studies; **Level D**, biomarkers that show plausible therapeutic significance based on preclinical studies or

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MC-7414

Case ID	:	2600135672	Sample Type	:	BLOOD IN CCF DNA TUBE
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	may assist disease diagnosis and/or prognosis themselves or along with other biomarkers based on small studies or multiple case reports with no consensus.
Tier III: Variants of unknown clinical significance	Not observed at a significant allele frequency in the general or specific subpopulation or pancancer or tumor specific variant databases. No convincing published evidence of cancer Association

Test Methodology:

- The cell-free DNA (cell-free total nucleic acid; cfTNA) isolated from the plasma fraction of whole blood.
- A single pool of multiplex PCR primers used to preparation of libraries from cell-free DNA (cfDNA) obtained from the plasma fraction of a single tube of whole blood. In process quality controls were determined for prepared library.
- The libraries were sequenced at minimum mean depth: >**25,000x** on Illumina Novaseq 6000 next generation sequencing platform.
- Highly automated workflow and streamlined bioinformatics analysis pipeline were used to provide a SNVs, Indels, CNVs and as part of the streamlined bioinformatics workflow, the mutational signature plot were generated.

Table: LiquidSeq- Colorectal Cancer Panel assay gene list.

MIER3	POLD1	TGFBR2	FZD3	AXIN2	CTNNA1
MLH1	POLE	TP53	GALNT12	BAX	
MLH3	PTEN	WBSCR17	GPC6	BLM	
MSH2	PTPN12	CTNNB1	GREM1	BMPR1A	
MSH3	RET	DCC	KIT	BRAF	
MSH6	RPS20	DMD	KRAS	BRCA1	
MUTYH	SCG5	EGFR	MAP2K4	BRCA2	
MYO1B	SLC9A9	ENG	MAP7	BUB1B	
NRAS	SMAD2	EP300	MET	CASP8	
PALB2	SMAD4	EPCAM	ACVR1B	CDC27	
PIK3CA	SRC	ERBB2	AKT1	CDH1	
PIK3R1	STK11	FBXW7	APC	CDK4	
PMS1	TCERG1	FGFR3	ATM	CDKN2A	
PMS2	TCF7L2	FLCN	ATP6V0D2	CHEK2	

MSI	% of Unstable Loci
MSI - Stable	0-19%

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MSI - Low / Intermediate	20-30%
MSI - High	>20%

Disclaimer:

- This test is limited to listed genes analysis only, It should be noted that this test does not sequence all bases in a human genome, not all variants have been identified or interpreted, and this report is limited only to variants with evidence for causing or contributing to disease/clinical details provided to SN Gene Lab Pvt Ltd.
- This assay is not meant to interrogate most promoter regions, deep intronic regions, or other regulatory elements, and does not detect single or multi-exon deletions or duplications.
- Due to poor quality of cell-free total nucleic acid; cfTNA possibility of assay failure/compromised results that include low gene coverage and low variant depth, cannot be ruled out
- No Guarantee: By providing drug and clinical trial information for the reported diagnosis, SN GeneLab Pvt. Ltd. is not guaranteeing that any drug or clinical trial is necessarily appropriate for this patient. Healthcare providers should evaluate and interpret the information provided in this report, along with all other available clinical information about this patient, to determine the best treatment decisions in their own independent medical judgment. Patient management decisions should not be based on a single test, including this one, nor solely on the information contained in this report.
- Drugs: The drugs listed on the report are not ranked in any specific order as to predicted efficacy or appropriateness for this patient. SN GeneLab Pvt. Ltd. makes no guarantee or promise as to the effectiveness or suitability (or lack thereof) of any drug listed on this report. For more detailed information, healthcare providers should refer to the package insert for each FDA-approved drug listed in this report, and go to clinicaltrials.gov for information regarding drugs in clinical trials.

***References*:**

1. Li, MM, et al. Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists. Journal of Molecular Diagnostics. 19.1 (2017): 4-23

2. N. Rizvi et al. Mutational landscape determines sensitivity to PD-1 blockade in non–small cell lung cancer. Science, 2015.

3. E. M. Van Allen et al. Genomic correlates of response to CTLA-4 blockade in metastatic melanoma. Science, 2015.

4. Meyer, LR, et al. The UCSC Genome Browser database: extensions and updates 2013. Nucleic Acids Research, (2013). 41 (Databaseissue): D64-9.

5. Wilm A. et al. LoFreq: A sequence-quality aware, ultra-sensitive variant caller for uncovering cell-population heterogeneity from high-throughput sequencing datasets. Nucleic Acids Res., 2012. 40(22): 11189-11201.

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6. Li H. et al. The Sequence Alignment/Map format and SAM tools. Bioinformatics, 2009. 25(16): 2078-9.

7. McLaren W. et al. The Ensembl Variant Effect Predictor. Genome Biology, 2016. 17(1): 122.

8. Daniel R.Z. et al. Ensembl 2018, Nucleic Acids Res., 2018. 46(D1): D754-D761.

9. The 1000 Genomes Project Consortium. A Global Reference for Human Genetic Variation. Nature. 526.7571 (2015): 68-74.

10. Lek, M, et al. Analysis of Protein-Coding Genetic Variation in 60,706 Humans. Nature. 536.7616 (2016): 285-91.

11. Schwarz JM, Cooper DN, Schuelke M and Seelow D. Mutationtaster2: mutation prediction for the deep-sequencing age. Nature Methods.2014; 11: 361-62.

12. R. Chaudhary et al. A scalable solution for tumor mutational burden from formalin-fixed, paraffin-embedded samples using the Oncomine Tumor Mutation Load Assay. Translational lung cancer research, 2018.

13. Li, H, and R, Durbin. Fast and accurate long-read alignment with Burrows-Wheeler transform. Bioinformatics. 26.5: (2010): 589-95.

------------------ End Of Report ------------------

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